

DNA and Meiosis — Complete Notes

Grade 12

Life Sciences

CAPS

What you'll learn

- the structure of DNA and how it stores genetic information
- how DNA makes an exact copy of itself (replication)
- the structure of RNA and the three main types of RNA
- how proteins are built from the instructions in DNA
- the genetic code and how codons work
- the process of meiosis and how it creates genetic variation
- how meiosis differs from mitosis
- how to answer DNA and meiosis questions using the correct terminology

DNA — the molecule of inheritance

DNA stands for deoxyribonucleic acid. It is the molecule that carries the genetic instructions used to build and run every living organism. In eukaryotic cells, most DNA is found in the nucleus, where it is wound around proteins called histones to form chromatin; chromatin condenses into chromosomes before cell division. Small amounts of DNA are also found in mitochondria and in the chloroplasts of plant cells.

The nucleotide — the monomer of DNA

A nucleotide is the basic building block (monomer) of a nucleic acid. Every DNA nucleotide is made of three parts:

1. a phosphate group
2. a five-carbon sugar called deoxyribose
3. a nitrogenous base

phosphate -- deoxyribose -- nitrogenous base (A, T, C or G)

There are four nitrogenous bases in DNA. Adenine and guanine are purines (double ring); thymine and cytosine are pyrimidines (single ring).

Base	Type	Pairs with	Hydrogen bonds
Adenine (A)	Purine	Thymine (T)	2
Thymine (T)	Pyrimidine	Adenine (A)	2
Cytosine (C)	Pyrimidine	Guanine (G)	3
Guanine (G)	Purine	Cytosine (C)	3

The double helix and complementary base pairing

Two strands of DNA twist around each other to form a double helix, first described by James Watson and Francis Crick in 1953 using the X-ray diffraction work of Rosalind Franklin and Maurice Wilkins. The two strands run in opposite directions — they are antiparallel — and are held together by weak hydrogen bonds

between complementary bases. Complementary base pairing means adenine always pairs with thymine, and cytosine always pairs with guanine. This is the base-pairing rule.

```
5'  A - T - G - C - C - T - A - G  3'
      :   :   :   :   :   :   :
3'  T - A - C - G - G - A - T - C  5'
```

Each ':' stands for the hydrogen bonds between one base pair (2 for A–T, 3 for C–G). The strong sugar-phosphate backbone runs along the outside; the bases point inward. Because the sequence of bases can differ from gene to gene, DNA can store enormous amounts of information — this base sequence is the genetic code.

Definition — complementary: describes bases that pair with each other in DNA (A with T, C with G), or a strand that is built to match another strand.

DNA replication — making an exact copy

Before a cell divides, it must copy its DNA so that each daughter cell receives a full set of instructions. DNA replication is semi-conservative: each new DNA molecule consists of one original strand and one newly built strand.

Where: in the nucleus, during interphase of the cell cycle (the S phase).

The steps are as follows:

1. The enzyme helicase unwinds the double helix and breaks the hydrogen bonds between the bases, so the two strands separate.
2. Each exposed strand acts as a template.
3. Free nucleotides in the nucleoplasm attach to the exposed bases by complementary base pairing (A to T, C to G).
4. The enzyme DNA polymerase joins the new nucleotides to each other, forming the new sugar-phosphate backbone.
5. Two identical DNA molecules are produced, each with one old strand and one new strand — hence semi-conservative.

```
Original DNA molecule
[ old strand 1 ] : : : : [ old strand 2 ]
      |
      DNA replication
      |
      v
Molecule A                Molecule B
[ old strand 1 ] : : : : [ new strand ]   [ new strand ] : : : : [ old strand 2 ]
```

RNA and its three types

RNA stands for ribonucleic acid. It is usually a single-stranded nucleic acid. RNA nucleotides contain ribose sugar (not deoxyribose) and the base uracil (U) in place of thymine.

Feature	DNA	RNA
Sugar	Deoxyribose	Ribose
Strands	Double helix	Usually single strand
Bases	A, T, C, G	A, U, C, G
Location	Mainly the nucleus	Nucleus and cytoplasm
Function	Stores genetic information	Carries and uses the code to make proteins

- mRNA (messenger RNA): a copy of a gene made during transcription. It carries the code from the nucleus to the ribosome. Each group of three bases on mRNA is a codon.
- tRNA (transfer RNA): carries a specific amino acid to the ribosome. It has an anticodon of three bases that pairs with the mRNA codon.
- rRNA (ribosomal RNA): forms part of the structure of a ribosome.

Protein synthesis

Protein synthesis happens in two stages: transcription and translation. A gene is a section of DNA that codes for one protein (polypeptide).

Step 1 — transcription (in the nucleus)

1. Part of the DNA unwinds and the hydrogen bonds break, exposing the bases of one strand (the template strand).
2. Free RNA nucleotides pair with the exposed bases by complementary base pairing. Because RNA has uracil, adenine on DNA pairs with uracil on mRNA (A to U), thymine pairs with adenine (T to A), cytosine pairs with guanine (C to G) and guanine pairs with cytosine (G to C).
3. The enzyme RNA polymerase joins the RNA nucleotides into a single strand of mRNA.
4. The mRNA detaches and the two DNA strands rejoin.
5. The mRNA leaves the nucleus through a nuclear pore and moves to a ribosome in the cytoplasm.

Step 2 — translation (at the ribosome)

1. The mRNA attaches to a ribosome.
2. A tRNA molecule with the matching anticodon brings the correct amino acid to the mRNA codon.
3. Amino acids are joined by peptide bonds to form a growing polypeptide chain.
4. The tRNA leaves to collect another amino acid.
5. Translation continues until a stop codon is reached; the polypeptide is released and folds into its functional shape (a protein).

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DNA (in the nucleus)
  |
  | transcription
  v
mRNA ---> nuclear pore ---> ribosome
                                |
                                | translation
                                v
                                polypeptide -> protein

```

The genetic code

- A codon is a sequence of three consecutive bases on mRNA that codes for one amino acid.
- It is a triplet code: three bases = one amino acid.
- There are four bases, so there are $4 \times 4 \times 4 = 64$ possible codons.
- The code is degenerate: more than one codon can code for the same amino acid.
- Some codons are signals. AUG is the start codon (it codes for methionine and starts translation). UAA, UAG and UGA are stop codons that end translation.
- The code is almost universal: the same codons usually code for the same amino acids in almost all organisms.

Meiosis

Definition — meiosis: a type of cell division that reduces the chromosome number from diploid ($2n$) to haploid (n) and produces four genetically different gametes.

Human body cells are diploid: $2n = 46$ chromosomes (23 pairs). Gametes (sperm and egg) are haploid: $n = 23$. Before meiosis begins, the DNA replicates during interphase, so each chromosome becomes double-stranded (two sister chromatids joined at a centromere).

Meiosis I

- Prophase I: homologous chromosomes (a matching pair, one from each parent) come together to form a bivalent (tetrad). Crossing over occurs — non-sister chromatids exchange pieces of DNA at points called chiasmata. This produces new combinations of alleles and is a source of genetic variation.
- Metaphase I: the bivalents line up on the equator of the cell. The way each pair lines up is random (independent assortment) — another source of variation.
- Anaphase I: the homologous chromosomes are pulled apart to opposite poles. The sister chromatids stay together.
- Telophase I and cytokinesis: two haploid cells form.

Meiosis II (similar to mitosis)

- Prophase II and Metaphase II: the chromosomes line up singly on the equator.
- Anaphase II: the sister chromatids are pulled apart to opposite poles.
- Telophase II and cytokinesis: four haploid cells form, each genetically unique.

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1 diploid cell (2n)
  |
  | meiosis I (homologous chromosomes separate)
  v
2 haploid cells (n) ---> meiosis II (sister chromatids separate) ---> 4 haploid cells

```

Significance of meiosis

1. It halves the chromosome number so that when a sperm (n) fertilises an egg (n), the zygote is diploid (2n) again — keeping the chromosome number constant from generation to generation.
2. It produces genetic variation through crossing over, independent assortment and random fertilisation.

Mitosis compared with meiosis

Feature	Mitosis	Meiosis
Number of divisions	1	2
Daughter cells produced	2	4
Chromosome number	Same as parent (diploid to diploid)	Halved (diploid to haploid)
Genetic variation	Daughter cells identical to parent	Daughter cells genetically different
Where it happens	Body (somatic) cells	Reproductive organs (gonads)
Function	Growth and repair; asexual reproduction	Produces gametes

Worked examples

Worked example 1 — from a DNA template to a protein

A section of the template strand of DNA has the base sequence 3' TAC CGT ATT 5'.

(a) Write the sequence of the mRNA made from this template. (b) Use these codes to name the amino acids added: AUG = methionine (start), GCA = alanine, UAA = stop.

Step 1 — apply complementary base pairing. mRNA is antiparallel to the template, A pairs with U, and RNA has no thymine.

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Template:  3' T A C C G T A T T 5'
mRNA:      5' A U G G C A U A A 3'

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Step 2 — read the mRNA in codons of three bases: AUG | GCA | UAA.

- AUG = methionine (start codon)
- GCA = alanine
- UAA = stop codon

Answer: the mRNA is 5' AUG GCA UAA 3'. The ribosome adds methionine and then alanine, and then translation stops, giving a dipeptide (Met–Ala).

Worked example 2 — complementary strand and hydrogen bonds

A single strand of DNA has the sequence 5' ATGCCTAG 3'.

(a) Write the sequence of the complementary strand. (b) How many hydrogen bonds hold the two strands together?

Step 1 — build the complementary strand. The strands are antiparallel and pairing is A with T and C with G.

```
Strand 1:  5' A  T  G  C  C  T  A  G  3'
Strand 2:  3' T  A  C  G  G  A  T  C  5'
```

Step 2 — count using 2 hydrogen bonds per A–T pair and 3 per C–G pair. The base pairs from left to right are: A–T (2), T–A (2), G–C (3), C–G (3), C–G (3), T–A (2), A–T (2), G–C (3).

Step 3 — add them: $2 + 2 + 3 + 3 + 3 + 2 + 2 + 3 = 20$.

Answer: the complementary strand is 3' TACGGATC 5', and the two strands are joined by 20 hydrogen bonds.

Worked example 3 — chromosome numbers through meiosis

A certain organism has body cells with 6 chromosomes ($2n = 6$).

(a) How many chromosomes are in each cell after meiosis I? (b) How many cells form after meiosis II, and how many chromosomes does each contain?

Step 1 — interphase: the DNA replicates. There are still 6 chromosomes, but each now has 2 sister chromatids (12 chromatids in total).

Step 2 — meiosis I (homologous pairs separate): the cell divides into 2 haploid cells. Each cell has 3 chromosomes, and each chromosome still has 2 sister chromatids.

Step 3 — meiosis II (sister chromatids separate): each of the 2 cells divides, giving 4 haploid cells. Each cell has 3 chromosomes, now single-stranded.

Answers: (a) 3 chromosomes per cell; (b) 4 cells, each containing 3 chromosomes.

Common mistakes

- Confusing the template strand with the coding strand. mRNA is complementary to the template, so a T on the template becomes A, and an A on the template becomes U.
- Writing thymine (T) in an mRNA sequence. RNA uses uracil (U) instead of thymine.
- Saying DNA replication is conservative. It is semi-conservative: each new molecule keeps one original strand.
- Mixing up the two divisions of meiosis. Homologous chromosomes separate in meiosis I; sister chromatids separate in meiosis II.
- Forgetting that crossing over happens in prophase I, between non-sister chromatids of homologous chromosomes.
- Saying meiosis produces two cells. It produces four.
- Writing that mitosis halves the chromosome number — it keeps it the same.
- Drawing meiosis with chromosomes that have not replicated. Replication happens in interphase before meiosis begins.
- Mixing up codon and anticodon: the codon is on mRNA, the anticodon is on tRNA.

Quick summary / cheat sheet

- DNA monomer = nucleotide = phosphate + deoxyribose sugar + nitrogenous base.
- Base pairing: A–T has 2 hydrogen bonds; C–G has 3 hydrogen bonds.
- Purines: adenine and guanine. Pyrimidines: thymine and cytosine.
- DNA replication is semi-conservative; helicase unwinds, DNA polymerase builds the new strands.
- RNA bases are A, U, C and G; RNA contains ribose sugar.
- Transcription (nucleus) makes mRNA from DNA. Translation (ribosome) makes protein from mRNA.
- A codon is three mRNA bases and codes for one amino acid. AUG is the start; UAA, UAG and UGA are stops.
- Meiosis takes 1 diploid cell to 4 haploid, genetically different cells.
- Variation comes from crossing over (prophase I), independent assortment (metaphase I) and random fertilisation.
- Meiosis I separates homologous chromosomes; meiosis II separates sister chromatids.

Practice questions

1. Give the complementary DNA sequence for the strand 5' GGATC 3', and state the number of hydrogen bonds.
2. Name the enzyme that unwinds the DNA double helix during replication.
3. During which phase of the cell cycle does DNA replication take place?
4. A DNA template strand reads 3' TAC GGT 5'. Write the mRNA sequence produced from it.
5. Explain what is meant by a semi-conservative method of replication.
6. Give TWO ways in which meiosis produces genetic variation.
7. State the number of chromosomes in a human haploid cell.
8. Explain why crossing over increases genetic variation.

Answers

1. The complementary strand is 3' CCTAG 5'; the pairs are G–C, G–C, A–T, T–A and C–G, giving 3 + 3 + 2 + 2 + 3 = 13 hydrogen bonds.
2. Helicase unwinds the double helix and breaks the hydrogen bonds between the bases.
3. DNA replication takes place in the nucleus during interphase, in the S phase of the cell cycle.
4. The template 3' TAC GGT 5' gives mRNA 5' AUG CCA 3' (T to A, A to U, C to G, G to C, G to C, T to A).
5. Semi-conservative means each new DNA molecule contains one original (parent) strand and one newly built strand, so half of the original molecule is conserved in every copy.
6. Any TWO of these: crossing over in prophase I, independent assortment of homologous chromosomes in metaphase I, and random fertilisation of gametes.
7. 23 chromosomes.
8. During crossing over, non-sister chromatids of homologous chromosomes exchange sections of DNA, creating new combinations of alleles that were not present in either parent, so the gametes formed are genetically different.

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